PROJECT PROPOSAL

# Olist E-Commerce Sales & Customer Experience Analytics

A data analytics project using the Brazilian Olist public dataset

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| **Team** | InsightHub |
| **Repository** | olist-ecommerce-analytics |
| **Tools** | Power BI, Power Query, DAX |
| **Dashboard plan** | Four dashboards, with one primary dashboard owner per assigned team member |
| **Date** | October 2026 |

This proposal defines the business questions, analytical scope, and four-dashboard structure. Findings will be reported only after data preparation, model validation, and analysis.

# 1. Project Overview

The project will analyze Olist's public Brazilian e-commerce dataset to explore order and sales trends, product/category performance, customer reviews and delivery outcomes, and seller/payment patterns. The team will prepare a documented Power BI data model and build four interactive dashboard pages, each with a distinct analytical purpose.

# 2. Problem Statement

Order volume alone does not fully describe e-commerce performance. Product mix, delivery timeliness, customer review outcomes, seller contribution, and payment behavior provide complementary views of the order journey. This project will measure patterns supported by the available records and avoid claiming causal relationships that the dataset cannot establish.

# 3. Project Objectives

- Measure order activity and item sales trends over the available period.
- Compare products and categories by recorded item sales value, order-item volume, and freight charges.
- Analyze review scores and examine their observed relationship with delivery timing and product/customer groups.
- Measure delivery duration and late-delivery rate using defined rules for eligible delivered orders.
- Compare seller contribution and explore payment types, installment counts, and payment values.
- Deliver four interactive Power BI dashboards with validated metrics, filters, and documented limitations.

# 4. Dataset and Feasibility

The project dataset comprises nine Olist CSV tables identified by the team: Customers, Geolocation, Order Items, Order Payments, Order Reviews, Orders, Products, Product Category Translation, and Sellers. Available fields include order timestamps/statuses, customer and seller identifiers/locations, item prices and freight values, product categories and attributes, review scores, and payment types/installments/values.

The listed tables do not establish all possible business metrics. Profit margin, marketing attribution, inventory availability, discount effectiveness, carrier-level performance, and actual-versus-target comparisons are not committed deliverables unless valid supporting fields or additional approved data are identified. The team will profile data completeness and confirm the observed time range before finalizing measures.

# 5. Four-Dashboard Structure and Business Questions

The four dashboards below are designed to divide the analysis into clear workstreams. Questions and candidate measures will be finalized after data profiling, relationship validation, and KPI definition.

## Dashboard 1 — Executive Sales & Order Trends

Purpose: provide an overview of order activity, recorded item sales value, and geographic/category contribution.

- How do distinct order counts and item sales value change by month and year in the available dataset period?
- Which months or periods have the highest and lowest order activity?
- How is recorded item sales value distributed across customer states?
- What share of item sales value comes from each product category?
- How do order-status inclusion rules affect the interpretation of the headline KPIs?

**Data/measure notes:** Candidate measures: distinct orders, item sales value (sum of order-item prices), order-item count, freight value, and average item sales value per order after validating definitions.

## Dashboard 2 — Product & Category Performance

Purpose: compare product mix and identify categories/products that contribute most to recorded item sales and freight.

- Which product categories contribute the largest and smallest shares of item sales value?
- Which products lead by item sales value and by order-item count, and do the rankings differ?
- Which categories have relatively high freight charges compared with item sales value?
- Where available product weight/dimension fields are populated, how do these attributes vary by category and relate descriptively to freight charges?
- Which categories show high item sales value but comparatively low order-item volume, or the reverse?

**Data/measure notes:** Fields: product/category, item price, freight value, product weight and dimensions. Missing product attributes will be disclosed; sales value is not profit.

## Dashboard 3 — Customer Experience & Delivery Performance

Purpose: combine customer review outcomes with delivery timeliness to explore patterns in the post-purchase experience.

- How are review scores distributed, and what share of submitted scores fall into documented low, middle, and high groups?
- What proportion of eligible delivered orders arrived after the estimated delivery date?
- Do orders delivered after the estimated date show different review scores from orders delivered on or before the estimate?
- How do delivery duration and review scores vary by product category and customer state?
- Which geographic or category groups combine substantial review volume with a relatively high share of low scores or late deliveries?

**Data/measure notes:** Fields: review score/ID, order ID, category, customer/seller locations, purchase/delivery/estimated-delivery timestamps. Canceled, unavailable, or not-yet-delivered orders will not be silently treated as delivered orders. Results show associations, not proof of causation.

## Dashboard 4 — Seller & Payment Insights

Purpose: examine seller contribution and the recorded mix of payment behavior.

- Which sellers contribute the largest item sales value and order-item volume?
- How do seller locations differ in recorded sales contribution and delivery/review indicators?
- What is the distribution of payment types, and how does it vary over time?
- How do payment installments and recorded payment values vary across orders?
- How do seller-level review or delivery indicators differ when there are enough observations for a fair comparison?

**Data/measure notes:** Payment records may contain multiple rows per order. Payment value and item sales must be analyzed separately to avoid duplicated totals. Seller comparisons will use sample-size checks.

# 6. Scope

**In scope**

- Data profiling, cleaning, and transformation with Power Query.
- A relational model with documented table grain, join keys, and validated relationships.
- Transparent KPI definitions and DAX measures where appropriate.
- Four interactive Power BI dashboards with suitable slicers and cross-filtering; drill-through where it supports the questions.
- Evidence-based findings and recommendations, with limitations clearly stated.

### Out of scope unless valid supporting data is identified

- Causal claims that delivery delays directly cause low ratings.
- Profit margin, discount impact, marketing attribution, advertising ROI, inventory/stock analysis, or target attainment without required fields.
- Carrier-level service comparisons without carrier identifiers and route-level records.
- Forecasting or predictive modeling as a guaranteed deliverable; this would require a separate feasibility decision.

# 7. Tools and Analytical Approach

**Tools:** Power BI Desktop, Power Query, DAX, and GitHub for project documentation and collaboration.

- Profile missing values, duplicates, key uniqueness, timestamps, category coverage, and table grain.
- Avoid joins between fact tables that can multiply order-item or payment totals.
- Define order, item-sales, payment, review, and delivery KPIs with explicit eligibility rules.
- Validate totals and sample records before interpreting charts.
- Record assumptions and limitations; do not present planned questions as established findings.

# 8. Expected Deliverables

- Project Proposal, Project Plan/timeline, Task Assignment & Roles, Risk Assessment, and Project KPIs.
- Literature Review and Requirements Gathering documents following DEPI guidance.
- Documented cleaning workflow and data model.
- Power BI report with four agreed dashboard pages and interactive controls.
- Final summary of validated findings, recommendations, and limitations.

# 9. Team and Timeline

Team: InsightHub. The four dashboards will be assigned to four team members as primary owners. The team has six members overall; remaining members can support shared data preparation, data modeling, quality assurance, documentation, and integration. Individual names and exact responsibilities should be confirmed in the separate Task Assignment & Roles document.

The planning/documentation milestones for Project Planning & Management, Literature Review, and Requirements Gathering are scheduled for 16 October 2026 according to the program timeline reviewed by the team. Detailed task owners, dependencies, and milestones will be documented in the Project Plan.

# 10. Success Criteria

- Each dashboard answers documented questions that can be answered from validated dataset fields.
- KPIs, table grain, relationships, and inclusion rules are documented and checked.
- The four dashboards have distinct purposes and avoid unnecessary duplication of headline metrics.
- Filters and visual interactions behave as intended.
- Reported conclusions are reproducible and distinguish observed patterns from causal claims.

Proposal status: planned scope, not findings. No numerical business conclusions are asserted here. Final questions and measures will be confirmed after profiling and validation.

Dataset source note: add the original public dataset source and applicable license/attribution details to the final repository documentation.